Welcome to the Introduction to Machine Learning!

Machine Learning is a subset of Artificial Intelligence that allows systems to learn from data.

Supervised learning involves labeled datasets and includes algorithms like Linear Regression, Decision Trees, and Support Vector Machines.

Unsupervised learning works with unlabeled data. Examples include Clustering and Dimensionality Reduction.

Reinforcement learning is based on rewards and penalties and is often used in robotics and game AI.

Python is a popular language for machine learning, along with libraries like scikit-learn, TensorFlow, and PyTorch.

Training a model requires data preprocessing, feature engineering, and choosing the right evaluation metrics.

Cross-validation ensures that the model generalizes well to unseen data.

Overfitting happens when the model performs well on training data but poorly on test data.

Underfitting means the model is too simple and fails to capture patterns.

Thanks for reading!